WIMU report

Benchmarking data contamination for Audio-Language Models

Kajetan Rożej, Rafał Szczepaniak, Wojciech Zarzecki

Introduction

Benchmark contamination sits at the intersection of evaluation benchmarks and a model's training data. When a model has been trained on the same samples later used to evaluate it, those evaluations are no longer credible, because we cannot tell whether the model genuinely generalized to the task or simply memorized the answer.

This risk is especially hard to rule out in audio-language models, which are assembled through a multi-stage training process. During pre-training, the language backbone is exposed to vast text corpora and the audio feature extractor to long hours of audio, so benchmark material can enter the model well before any evaluation takes place. The risk does not end there: during post-training the model is adapted to operate on audio tokens, and contamination can be introduced at this stage as well. Despite this, contamination in audio-language models has not been systematically examined. In this work we adapt contamination and membership inference methods originally designed for large language models and vision-language models to audio-language models, that is, models that take text and audio as input and produce text as output.

Datasets

We used three different datasets in our experiments:

- **Clotho-v2 (link)** - a highly curated dataset designed specifically for audio captioning. It features 15- to 30-second audio clips sourced from Freesound, with each clip paired with five distinct, crowdsourced natural language captions (each caption was treated separately)
- **agkphysics/AudioSet (link) -** a balanced subset of full AudioSet dataset downloaded in March 2023 to prevent problems with Youtube data accessibility
- **AudioCaps (link) -** A large-scale audio captioning dataset built directly on top of the AudioSet collection. It provides human-written, natural language text descriptions for roughly 46,000 of the 10-second YouTube audio clips found in AudioSet**.**

in each case, the training set was used as a reference (since the models were confirmed to have been trained on this data), while the validation and test sets were used as benchmarks to check for data leakage

Models

In our experiments, we evaluated a total of four audio-language models from the open-source Audio Flamingo family developed by NVIDIA. The evaluated models are:

- **Audio Flamingo 2-0.5B:** a highly lightweight variant of the second-generation model,
- **Audio Flamingo 2-1.5B:** mid-sized version strikes a practical balance between computational efficiency and expert-level reasoning for non-speech sounds and music.
- **Audio Flamingo 2-3B:** the flagship model of the Audio Flamingo 2 series.

- **Audio Flamingo 3:** This fully open-source, third-generation model unifies speech, ambient sound, and music understanding, introducing advanced features like flexible "chain-of-thought" reasoning and long-context processing for audio up to 10 minutes in length.

Implementation

The project follows the well-known **Lightning + Hydra "template" layout** for ML research, adapted from training to an *evaluation / detection* pipeline. The repository is organized as follows:

- `configs/` — the entire Hydra configuration tree, expressed as `.yaml` files (one per model, method, dataset, logger, etc.). Experiments are assembled by composition rather than by code changes.
- `src/` — the main source code for experiments, with dedicated directories for:
  - `models/` — wrappers around concrete audio-language models (Audio Flamingo 2, Audio Flamingo 3),
  - `methods/` — contamination / MIA detection algorithms (Yeom perplexity, Min-K%, Min-K%++, VL-MIA, CoDeC, MM-Detect),
  - `data/` — dataset readers (Clotho, AudioCaps, AudioSet, and their MM-Detect variants).

  Each of these groups is built on top of an abstract base class (`BaseAudioLanguageModel`, `MethodBaseClass`, `BaseAudioTextDataset` / `BaseMMDetectDataset`) that defines a common interface for all of its members — with a few intentional exceptions where a method's logic does not fit the standard signature (e.g. MM-Detect operates at the dataset level rather than per sample).

  The `src/utils/` folder provides cross-cutting helpers (logging, Hydra utilities, W&B initialization), and there are three Hydra entrypoints, one per detection family:

  - `src/contamination.py` — CoDeC,
  - `src/eval_mia.py` — perplexity-based MIA methods (Yeom, Min-K%, Min-K%++, VL-MIA),
  - `src/eval_mm_detect.py` — MM-Detect.
- `scripts/` — auxiliary Python and Bash scripts for environment setup, dataset download and preprocessing, NLLB back-translation, and launching experiments on the Athena HPC cluster.

Thanks to this highly modular, open design, adding a new dataset or model and wiring it into the evaluation ecosystem is straightforward: all that is required is a wrapper class that implements the relevant common interface — no changes to the entrypoints, methods, or configuration loaders are needed.

For experiment tracking we used **Weights & Biases**. All runs log their configuration, per-sample scores, and aggregated metrics (ROC-AUC, best-threshold accuracy, CR / PCR / Δ for MM-Detect), which makes it easy to manage, compare, filter, and visualize results as new experiments come in.

Benchmarking

idea of benchmark contamination - looking at outcome, stress testing model generalization capabilities

benchmark contamination for LLMs methods and their adaptations - theory

- CoDeC
- MM-Detect

Alternative approach MIAs - looking at cause instead of outcome - literal training data detection

- Perplexity (baseline)
- Min-K% and Min-K%++

- VL-MIA

CoDeC

CoDeC (Detecting Data Contamination in LLMs via In-Context Learning) was introduced by Zawalski et al. and targets dataset-level contamination detection in large language models. The core insight is that **in-context examples drawn from a memorized dataset provide no new information to the model**, whereas the same examples from genuinely unseen data improve predictions through in-context learning. This asymmetry is exploited to build a parameter-free, model-agnostic contamination signal.

The original method was designed exclusively for text-only LLMs. We adapted it to the audio-language setting by replacing plain text demonstrations with **(audio, text) pairs**, preserving the method's logic while introducing an additional modality as the grounding signal.

The adapted pipeline runs as follows for each target sample (a, t):

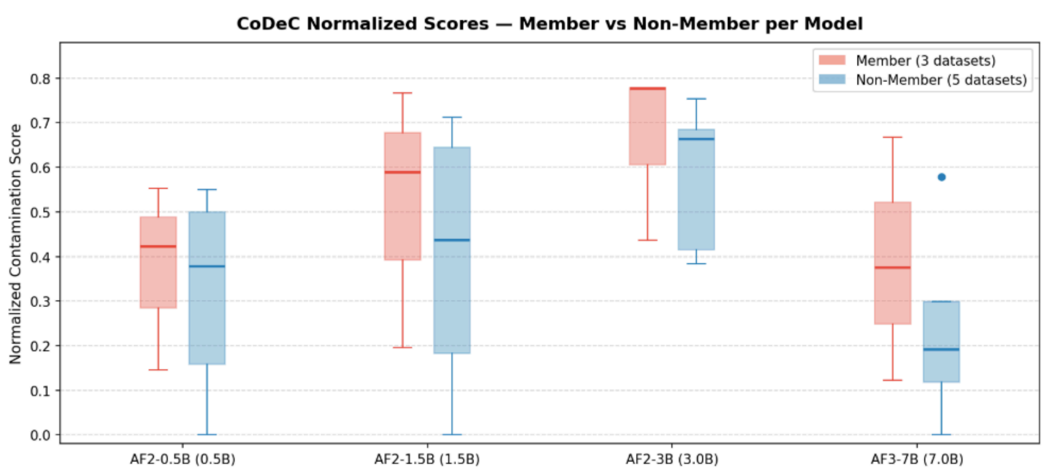
1. **Baseline scoring.** The model scores the target caption t conditioned on its paired audio a alone, producing a per-token log-likelihood sequence.
2. **Context pool construction.** A pool of P = 50 (audio, caption) pairs is sampled from the *member* split of the dataset under test. This pool is shared across all target samples within a run.
3. **In-context scoring.** N demonstrations are sampled from the pool (excluding the current target) and prepended to the query. The model scores t again, now conditioned on both the target audio a and the N context demonstrations.
4. **Per-sample delta.** The difference$\Delta(x) = \overline{\log p}_{\text{ctx}}(t \mid a) - \overline{\log p}_{\text{no-ctx}}(t \mid a)$ is computed over a trimmed token window (first 10 tokens dropped to avoid prompt-leakage artifacts, identical to the reference implementation).
5. **Dataset-level score.** The contamination score $S_{\text{CoDeC}}(D) = \frac{1}{N}\sum_i \mathbf{1}[\Delta(x_i) < 0]$ counts the fraction of samples for which adding context *reduced* confidence — the hallmark of a memorized dataset.

This table presents summary of all runs for CoDeC on two series of models and 5 datasets.

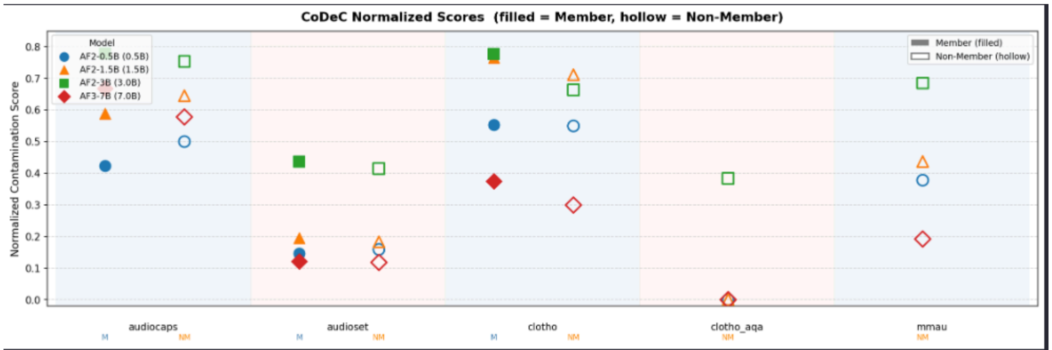
Model	Dataset	Member Norm Score	Non-Member Norm Score	n member	n non-member
audio-flamingo-2-0.5B	audiocaps	0.423	0.5	1000	1000
audio-flamingo-2-0.5B	audioset	0.146	0.159	1000	1000
audio-flamingo-2-0.5B	clotho	0.553	0.5494	1000	648
audio-flamingo-2-0.5B	clotho_aqa	N/A	0.0	-	1000
audio-flamingo-2-0.5B	mmau	N/A	0.378	-	1000
audio-flamingo-2-1.5B	audiocaps	0.588	0.645	1000	1000
audio-flamingo-2-1.5B	audioset	0.195	0.183	1000	1000
audio-flamingo-2-1.5B	clotho	0.766	0.7114	1000	648

Model	Dataset	Member Norm Score	Non-Member Norm Score	n member	n non-member
audio-flamingo-2-1.5B	clotho_aqa	N/A	0.001	-	1000
audio-flamingo-2-1.5B	mmau	N/A	0.437	-	1000
audio-flamingo-2-3B	audiocaps	0.778	0.753	1000	1000
audio-flamingo-2-3B	audioset	0.436	0.415	1000	1000
audio-flamingo-2-3B	clotho	0.777	0.6636	1000	648
audio-flamingo-2-3B	clotho_aqa	N/A	0.384	-	1000
audio-flamingo-2-3B	mmau	N/A	0.685	-	1000
audio-flamingo-3-hf	audiocaps	0.668	0.578	1000	1000
audio-flamingo-3-hf	audioset	0.122	0.118	1000	1000
audio-flamingo-3-hf	clotho	0.375	0.2994	1000	648
audio-flamingo-3-hf	clotho_aqa	N/A	0.0	-	1000
audio-flamingo-3-hf	mmau	N/A	0.192	-	1000

The box plots below aggregate normalized contamination scores across datasets for each model, comparing training members (three datasets) against non-members (five datasets). Member scores are consistently higher than non-member scores across all four model versions, suggesting that CoDeC transfers effectively from the text-only LLM setting to audio-language models.



The scatter plot below breaks scores down by individual dataset and model. Although the member–non-member separation is visible at the distribution level, the overlap between individual data points is substantial, indicating that the signal is too noisy to draw reliable conclusions about contamination for any single sample.



MM-Detect

The MM-Detect framework was described in the paper "Both Text and Images Leaked! A Systematic Analysis of Data Contamination in Multimodal LLM" by Song et al.. The main objective of the method is to detect data contamination—specifically, to quantify how much of a model's performance stems from the unintentional memorization of benchmark data rather than genuine generalization. To evaluate this, the authors suggest two primary approaches:

- **Option Order Sensitivity Test** for multiple-choice datasets.
- **Slot Guessing for Perturbed Captions** for free-text caption datasets.

Because of the nature of our datasets, we performed our experiments using the caption masking method. The workflow behind it can be described in a set of simple steps:

1. We take the original caption, translate it into a different intermediate language, and then back-translate it into the original language to generate a paraphrased sentence.
2. To ensure we mask a meaningful word, keyword extraction is performed, limiting candidates strictly to nouns, adjectives, and verbs.
3. One randomly selected keyword from both the original sentence and the back-translated sentence is masked independently.
4. We ask the model to predict the masked word based on the media context (audio, in our case) for both the original and paraphrased masked sentences.
5. We calculate the difference between the percentage of correct answers on the original dataset (Correct Rate) and the back-translated dataset (Perturbed Correct Rate). This performance reduction is represented as delta.

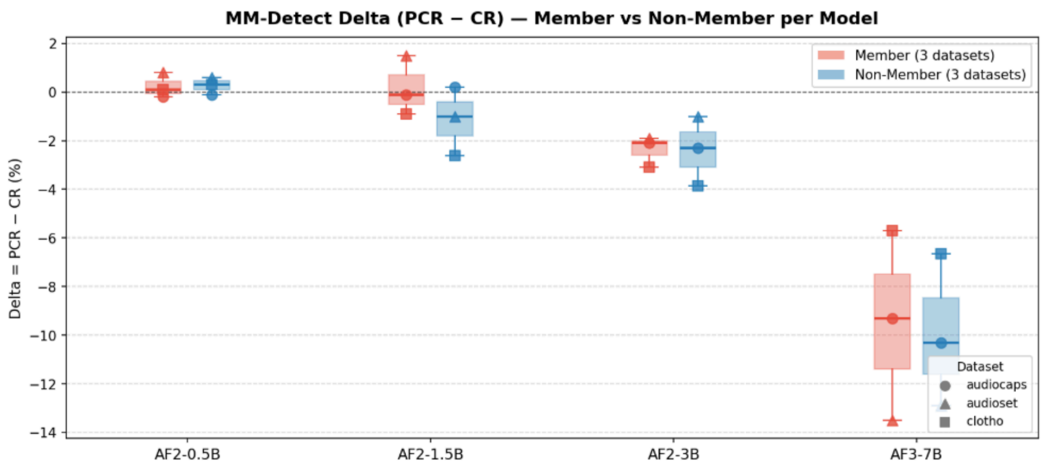
The intuition behind this method is that if a model relies on memorized training data, it will perform significantly better at predicting the missing word in the exact original sentence, but will fail on the semantic-equivalent back-translated paraphrase.

Model	Dataset	Member	CR	PCR	Delta	n
audio-flamingo-2-0.5B	audiocaps	No	0.7	0.6	-0.1	1000
audio-flamingo-2-0.5B	audiocaps	Yes	0.3	0.1	-0.2	1000
audio-flamingo-2-	audioset	No	0.4	1.0	0.6	1000

Model	Dataset	Member	CR	PCR	Delta	n
0.5B						
audio-flamingo-2-0.5B	audioset	Yes	0.3	1.1	0.8	1000
audio-flamingo-2-0.5B	clotho	No	0.463	0.7716	0.3086	648
audio-flamingo-2-0.5B	clotho	Yes	0.6	0.7	0.1	1000
audio-flamingo-2-1.5B	audiocaps	No	4.5	4.7	0.2	1000
audio-flamingo-2-1.5B	audiocaps	Yes	4.5	4.4	-0.1	1000
audio-flamingo-2-1.5B	audioset	No	1.8	0.8	-1.0	1000
audio-flamingo-2-1.5B	audioset	Yes	0.7	2.2	1.5	1000
audio-flamingo-2-1.5B	clotho	No	9.5679	6.9444	-2.6235	648
audio-flamingo-2-1.5B	clotho	Yes	9.1	8.2	-0.9	1000
audio-flamingo-2-3B	audiocaps	No	5.9	3.6	-2.3	1000
audio-flamingo-2-3B	audiocaps	Yes	6.5	4.4	-2.1	1000
audio-flamingo-2-3B	audioset	No	3.4	2.4	-1.0	1000
audio-flamingo-2-3B	audioset	Yes	5.0	3.1	-1.9	1000
audio-flamingo-2-3B	clotho	No	12.1914	8.3333	-3.858	648
audio-flamingo-2-3B	clotho	Yes	13.0	9.9	-3.1	1000
audio-flamingo-3-hf	audiocaps	No	36.5	26.2	-10.3	1000
audio-flamingo-3-hf	audiocaps	Yes	40.1	30.8	-9.3	1000
audio-flamingo-3-hf	audioset	No	29.7	16.8	-12.9	1000
audio-flamingo-3-hf	audioset	Yes	33.1	19.6	-13.5	1000
audio-flamingo-3-hf	clotho	No	29.0123	22.3765	-6.6358	648
audio-	clotho	Yes	36.4	30.7	-5.7	1000

Model	Dataset	Member	CR	PCR	Delta	n
flamingo-3-hf						

The box plot below aggregates delta values (PCR − CR) across the three evaluated datasets for each model, comparing training members against non-members.



MM-Detect predicts that contaminated models should score more negative Δ on members than non-members, since memorization favors the original wording over its paraphrase. The data does not show this: member and non-member Δ distributions overlap on every model, with means within ~1 point (e.g. AF3-7B: −9.5 vs −9.9), and in 7 of 12 (model, dataset) cells the non-member Δ is actually the more negative one. Variation is instead driven by model scale and dataset — AF3-7B sits at Δ ≈ −5 to −13 for both splits, AF2-0.5B near zero for both — consistent with paraphrase fragility rather than selective memorization, and matching Song et al.'s caveat that MM-Detect can mask contamination when models generalize over perturbations. As adapted to audio-language models, MM-Detect does not yield a usable contamination signal here.

MIA Methods

Membership Inference Attacks (MIAs) frame contamination detection as a per-sample classification problem: given a trained model and a candidate example, decide whether that example was part of the training set. The underlying assumption is that a model assigns systematically *higher* likelihood (lower loss) to data it has already seen than to fresh data drawn from the same distribution, because training explicitly minimizes loss on the training set. Each method below turns that intuition into a single scalar score per sample; we then sweep a threshold over those scores and report ROC-AUC against the ground-truth member / non-member labels. Unlike CoDeC and MM-Detect, MIAs operate at the **sample level** rather than the dataset level, which makes them strictly harder but also strictly more informative when they work.

In the audio-language setting, every method below scores output tokens conditioned by input pair (audio, prompt).

Perplexity

Perplexity is the simplest possible MIA baseline (Yeom et al. (2018)): the score for a sample is just the average negative log-likelihood (loss) the model assigns to the caption given the audio. Members are predicted to have lower loss, non-members higher loss. It uses no calibration, no reference model, and no token selection — any signal it produces comes entirely from the absolute confidence gap between seen and unseen captions.

Model	Dataset	n	ROC-AUC
audio-flamingo-2-0.5B	audiocaps	2000	0.4209
audio-flamingo-2-0.5B	audioset	2000	0.4726
audio-flamingo-2-0.5B	clotho	1648	0.5758
audio-flamingo-2-1.5B	audiocaps	2000	0.4234
audio-flamingo-2-1.5B	audioset	2000	0.4857
audio-flamingo-2-1.5B	clotho	1648	0.5855
audio-flamingo-2-3B	audiocaps	2000	0.4401
audio-flamingo-2-3B	audioset	2000	0.4807
audio-flamingo-2-3B	clotho	1648	0.6403
audio-flamingo-3-hf	audiocaps	2000	0.719
audio-flamingo-3-hf	audioset	2000	0.5059
audio-flamingo-3-hf	clotho	1648	0.7633

Min-K

Min-K% (Shi et al., 2024) refines the perplexity baseline by averaging the log-likelihood only over the *K%* of tokens with the **lowest** probability under the model. The intuition is that memorization should manifest most clearly on the "hard" tokens — rare or surprising words that an un-trained model would assign very low probability to but a model that has seen the exact caption can still predict confidently. Averaging over the bottom-K% suppresses noise from easy, high-frequency tokens that members and non-members both predict well.

Model	Dataset	n	ROC-AUC
audio-flamingo-2-0.5B	audiocaps	2000	0.4163
audio-flamingo-2-0.5B	audioset	2000	0.5186
audio-flamingo-2-0.5B	clotho	1648	0.5453
audio-flamingo-2-1.5B	audiocaps	2000	0.4102
audio-flamingo-2-1.5B	audioset	2000	0.5096
audio-flamingo-2-1.5B	clotho	1648	0.5682
audio-flamingo-2-3B	audiocaps	2000	0.425
audio-flamingo-2-3B	audioset	2000	0.498
audio-flamingo-2-3B	clotho	1648	0.6073
audio-flamingo-3-hf	audiocaps	2000	0.72
audio-flamingo-3-hf	audioset	2000	0.5065
audio-flamingo-3-hf	clotho	1648	0.7523

Min-K++

Min-K%++ (Zhang et al., 2024) keeps the bottom-K% token selection of Min-K% but replaces the raw log-probability with a **normalized** score: each token's log-probability is centered and scaled by the mean and standard deviation of the model's full next-token distribution at that position. This calibration removes the influence of overall distribution sharpness (e.g. low-entropy positions where any model is confident) and isolates how much the *specific* observed token stands out from its alternatives — a more discriminative signal of memorization than absolute likelihood.

Model	Dataset	n	ROC-AUC
audio-flamingo-2-0.5B	audiocaps	2000	0.4148
audio-flamingo-2-0.5B	audioset	2000	0.511
audio-flamingo-2-0.5B	clotho	1648	0.558
audio-flamingo-2-1.5B	audiocaps	2000	0.424
audio-flamingo-2-1.5B	audioset	2000	0.5161
audio-flamingo-2-1.5B	clotho	1648	0.569
audio-flamingo-2-3B	audiocaps	2000	0.4353
audio-flamingo-2-3B	audioset	2000	0.4943
audio-flamingo-2-3B	clotho	1648	0.6257
audio-flamingo-3-hf	audiocaps	2000	0.7392
audio-flamingo-3-hf	audioset	2000	0.4968
audio-flamingo-3-hf	clotho	1648	0.7092

VL-MIA Entropy

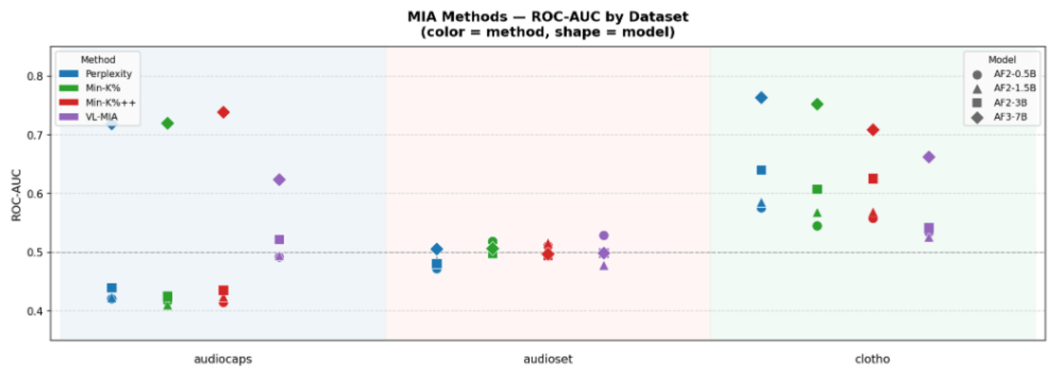
VL-MIA (Li et al., 2024) was designed for vision-language models and exploits a different signal than the loss-based attacks above: the **entropy** of the model's next-token distribution. The hypothesis is that on memorized samples the model is not just more accurate, but also more *certain* — its predictive distribution collapses around the observed token, while on unseen samples it stays diffuse. We use the entropy variant of VL-MIA, scoring each sample as the mean per-token entropy of $p(\cdot \mid \text{audio}, t_{<i})$; lower entropy indicates membership. Because it ignores the identity of the observed token entirely and only looks at the *shape* of the distribution, it captures memorization signals that loss-based scores can miss.

VL-MIA (Li et al., 2024) was designed for vision-language models and exploits a different signal than the loss-based attacks above: the **entropy** of the model's next-token distribution. The hypothesis is that on memorized samples the model is not just more accurate, but also more *certain* — its predictive distribution collapses around the observed token, while on unseen samples it stays diffuse. We use the entropy variant of VL-MIA, scoring each sample as the mean per-token entropy of $p(\cdot \mid \text{audio}, t_{<i})$; lower entropy indicates membership. Because it ignores the identity of the observed token entirely and only looks at the *shape* of the distribution, it captures memorization signals that loss-based scores can miss.

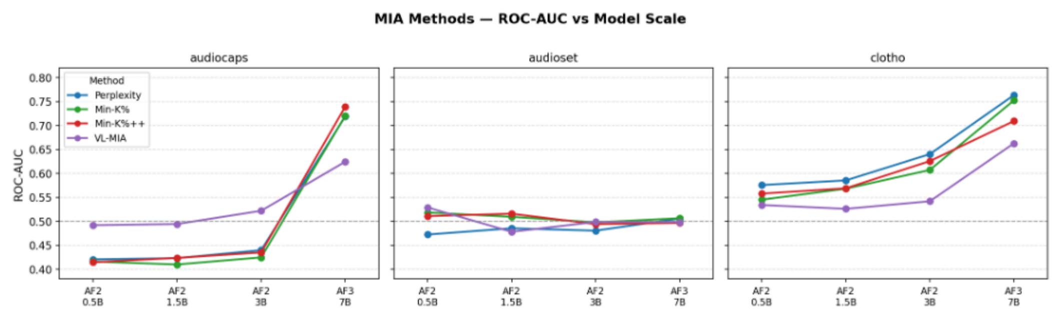
Model	Dataset	n	ROC-AUC
audio-flamingo-2-0.5B	audiocaps	2000	0.4921
audio-flamingo-2-0.5B	audioset	2000	0.5292
audio-flamingo-2-0.5B	clotho	1648	0.5342
audio-flamingo-2-1.5B	audiocaps	2000	0.4945
audio-flamingo-2-1.5B	audioset	2000	0.4783
audio-flamingo-2-1.5B	clotho	1648	0.5262
audio-flamingo-2-3B	audiocaps	2000	0.5222
audio-flamingo-2-3B	audioset	2000	0.4988
audio-flamingo-2-3B	clotho	1648	0.542
audio-flamingo-3-hf	audiocaps	2000	0.6244
audio-flamingo-3-hf	audioset	2000	0.4986

Model	Dataset	n	ROC-AUC
audio-flamingo-3-hf	clotho	1648	0.6627

The scatter plot below combines all five MIA methods into a single view. Points are grouped by dataset, colored by method, and shaped by model version. The dashed line marks random-chance performance (ROC-AUC = 0.5).



The line chart below shows how ROC-AUC evolves with model scale for each method and dataset.



Read together, the two plots tell a consistent story dominated by strong impact of model scale and dataset — and a much weaker of method choice.
On both AudioCaps and Clotho, every loss-based method climbs monotonically with model size and jumps sharply at AF3-7B (≈0.72–0.76 ROC-AUC), well above chance. The smaller AF2 variants (0.5B / 1.5B / 3B) sit close together, and the gap to AF3 is much larger than any gap between methods. This suggest that memorization is a property that emerges with capacity, and bigger checkpoints leak more. Clotho is approachable dataset for MIAs while 2 other remain to random guessing. No MIA we evaluated produces a sample-level membership decision reliable enough to call any individual caption "contaminated": even the best ROC-AUC (≈0.76) leaves a large false-positive rate at any usable recall, what matches observation for LLMs (Duan et. al 2024).

Conclusions

Across the methods we evaluated, CoDeC was the most reliable contamination detector for audio-language models, separating members from non-members consistently on the three benchmarks where both splits exist. Its practical limitation is that it operates at the dataset level and only becomes interpretable once a distribution of per-dataset scores is available: a single score in isolation is hard to act on, but an outlier within a comparable pool is a strong contamination signal. MM-Detect, by contrast, did not produce a membership-discriminative signal in our setting and we would not recommend it as a standalone audio contamination test. MIA methods do not give a sample-level guarantee, even on larger models, although ROC-AUC rises sharply at AF3-7B on AudioCaps and Clotho (≈0.72–0.76).

We suggest that a CoDeC score outlier together with above-chance MIA scores on the same dataset can be a strong premise for contamination.

References

- Zawalski, Michał, et al. "Detecting Data Contamination in LLMs via In-Context Learning." *arXiv preprint arXiv:2510.27055* (2025).
- Song, Dingjie, et al. "Both Text and Images Leaked! A Systematic Analysis of Data Contamination in Multimodal LLM." *arXiv preprint arXiv:2411.03823* (2024).
- Duan, Michael, et al. "Do membership inference attacks work on large language models?." *arXiv preprint arXiv:2402.07841* (2024).
- Yeom, Samuel, et al. "Privacy risk in machine learning: Analyzing the connection to overfitting." *2018 IEEE 31st computer security foundations symposium (CSF)*. IEEE, 2018.
- Shi, Weijia, et al. "Detecting pretraining data from large language models." *International Conference on Learning Representations*. Vol. 2024. 2024.
- Zhang, Jingyang, et al. "Min-K%++: Improved Baseline for Pre-Training Data Detection from Large Language Models." *International Conference on Learning Representations*. Vol. 2025. 2025.
- Li, Zhan, et al. "Membership inference attacks against large vision-language models." *Advances in Neural Information Processing Systems* 37 (2024): 98645-98674.